# QueryGenie

## A Self-Correcting, Confidence-Aware Natural Language to SQL Interface

STUDENTS | INNOVATION | RESEARCH | IMPACT

## Abstract

**QueryGenie** lets anyone ask a database questions in plain English. Text-to-SQL models answer every question with the same confidence, so a wrong answer looks exactly like a right one. We reproduce CodeS (SIGMOD 2024), an open-source text-to-SQL model, to 77.8% execution accuracy on Spider (paper: 77.9%), and add confidence-aware abstention: the system runs every candidate query the model writes and checks whether they agree. When they don't, it asks the user instead of guessing. It answers 90.4% of questions and is right on 83.3% of them, with a third fewer wrong answers — a statistically significant gain. It runs fully locally in a desktop app.

## Objectives

1. Reproduce CodeS-1B on Spider within the ±5–10% tolerance
2. Find which parts of the CodeS pipeline matter (ablation)
3. Make the system say when it is unsure — tested statistically
4. Measure accuracy on realistically worded questions
5. Deliver a desktop app that runs fully locally

## Key Innovations & Contributions

**1. Confidence-aware abstention**
Run all candidate queries, count how many agree; ask "Did you mean…?" when they don't

**2. Faithful, audited reproduction**
77.8% vs 77.9% with the official evaluator, after auditing our own harness

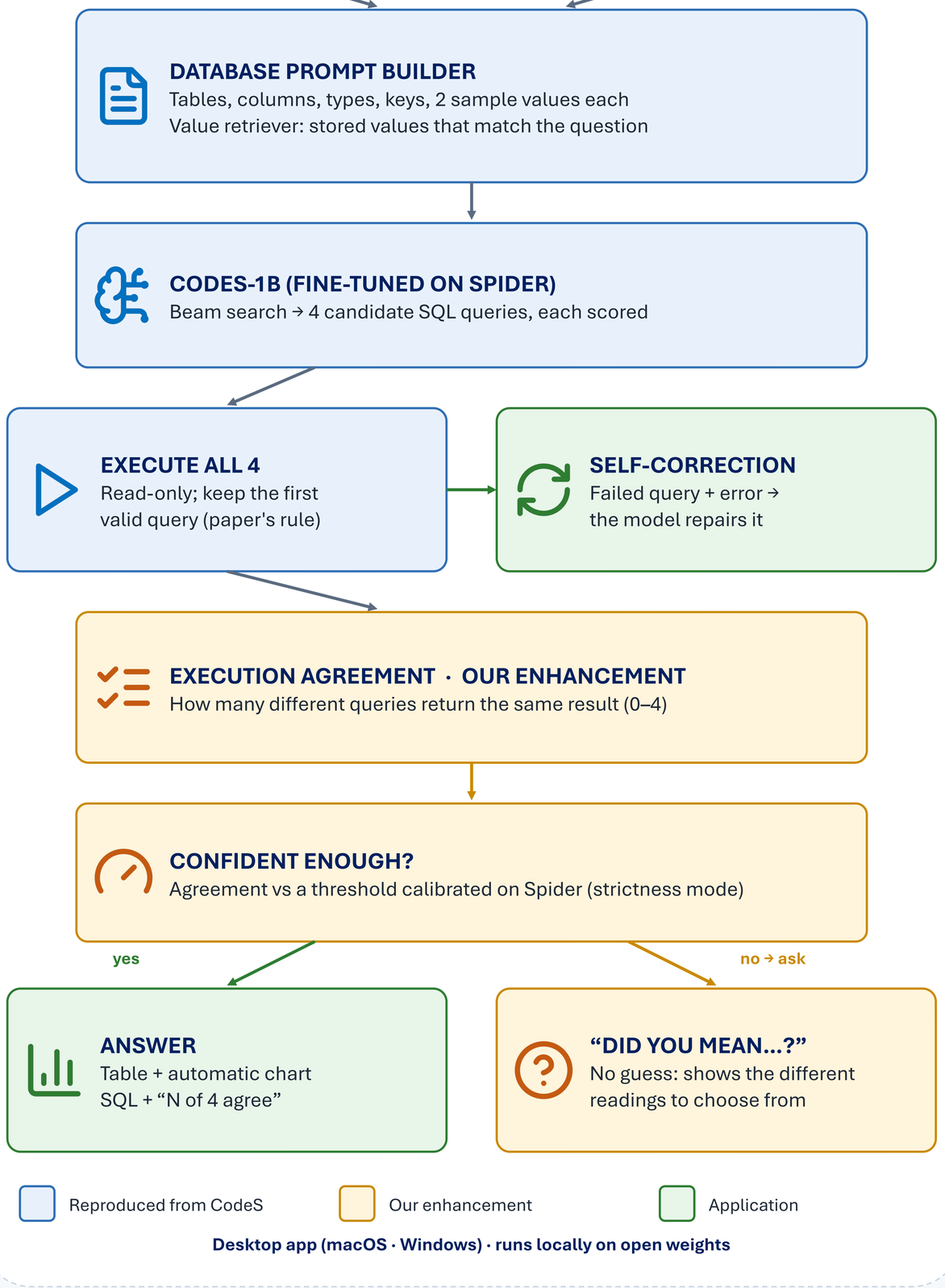
**3. Execution-guided self-correction**
A failed query goes back to the model with the database error

**4. Private by design**
Open weights, runs on the user's computer — no schema sent to a third party

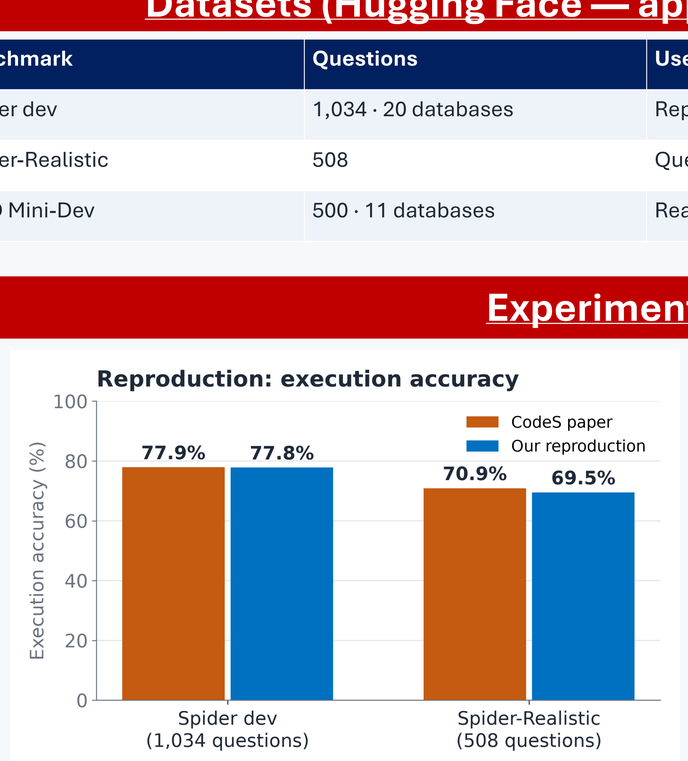
## Datasets (Hugging Face — approved source)

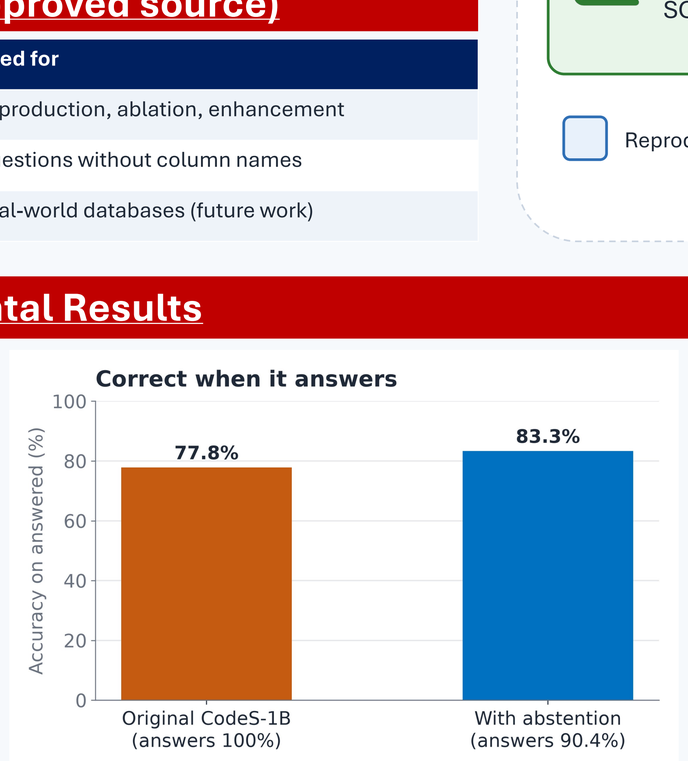
| Benchmark | Questions | Used for |
|---|---|---|
| Spider dev | 1,034 · 20 databases | Reproduction, ablation, enhancement |
| Spider-Realistic | 508 | Questions without column names |
| BIRD Mini-Dev | 500 · 11 databases | Real-world databases (future work) |

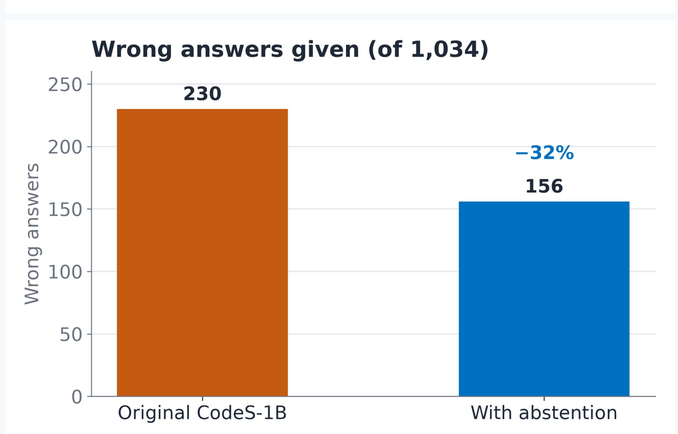
## Proposed Methodology / System Architecture

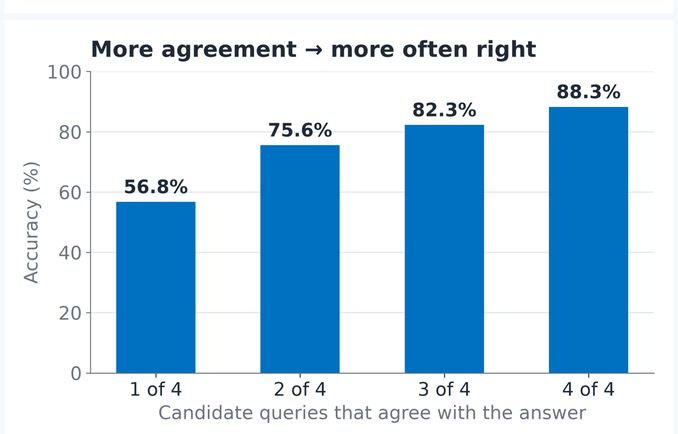



## Experimental Results




**Significant:** paired t-test $p = 7.7\times10^{-10}$, Wilcoxon $p = 9.8\times10^{-4}$, bootstrap 95% CI +4.5 to +6.7 points.
**Robust:** counting each different query only once, accuracy at 90% coverage is still 83.5%.

## References

**[1]** Li, H., et al. (2024). CodeS: Towards building open-source language models for text-to-SQL. arXiv:2402.16347.

**[2]** Yu, T., et al. (2018). Spider: A large-scale human-labeled dataset for complex and cross-domain semantic parsing and text-to-SQL task. arXiv:1809.08887.

**[3]** Deng, X., et al. (2020). Structure-grounded pretraining for text-to-SQL. arXiv:2010.12773.

**[4]** Li, J., et al. (2023). Can LLM already serve as a database interface? A big bench for large-scale database grounded text-to-SQLs. arXiv:2305.03111.

**[5]** Pourreza, M., & Rafiei, D. (2023). DIN-SQL: Decomposed in-context learning of text-to-SQL with self-correction. arXiv:2304.11015.

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**[7]** Geifman, Y., & El-Yaniv, R. (2017). Selective classification for deep neural networks. arXiv:1705.08500.

**[8]** Zhong, R., Yu, T., & Klein, D. (2020). Semantic evaluation for text-to-SQL with distilled test suites. arXiv:2010.02840.

## Team Details

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## Future Work

- CodeS-3B and more, more diverse candidate queries
- Real-world databases (BIRD)
- A platform for any database (Spark, Snowflake, NoSQL) with access control